Answer to the question no 5

(b)

of z_0 in which the point z_0 is omitted, i.e. $0 < |z - z_0| < \delta$.

2. **Limit Points.** A point z_0 is called a *limit point*, *cluster point*, or *point of accumulation* of a point set S if every deleted δ neighbourhood of z_0 contains points of S.

Since δ can be any positive number, it follows that S must have infinitely many points. Note that z_0 may or may not belong to the set S.

[illegible] limit point of S belongs to S, i.e. if S contains all its

called *compact*.

5. **Interior, Exterior and Boundary Points.** A point z_0 is called an *interior point* of a set S if we can find a δ neighbourhood of z_0 all of whose points belong to S. If every δ neighbourhood of z_0 contains points belonging to S and also points not belonging to S, then z_0 is called a *boundary point*. If a point is not an interior or boundary point of a set S, it is an *exterior point* of S.

10. [illegible] region.

11. **Regions.** If to an open region or domain we add some, all or none of its limit points, we obtain a set called a *region*. If all the limit points are added, the region is *closed*; if none are added, the region is *open*. In this book whenever we use the word *region* without qualifying it, we shall mean *open region* or *domain*.

12. **Union and Intersection of Sets.** A set consisting of all points belonging to set S_1 or set S_2 or [illegible]

Answer to the question no 6

(b)

2.30 Prove that $\lim_{z \to 0} \dfrac{\bar{z}}{z}$ does not exist.

Solution If the limit is to exis,t it must be independent of the manner in which z approaches the point 0. Let $z \to 0$ along the x axis. Then $y = 0$, and $z = x + iy = x$ and $\bar{z} = x - iy = x$, so that the required limit is

$$\lim_{x \to 0} \frac{x}{x} = 1$$

Let $z \to 0$ along the y axis. Then $x = 0$, and $z = x + iy = iy$ and $\bar{z} = x - iy = -iy$, so that the required limit is

$$\lim_{x \to 0} \frac{-iy}{iy} = -1$$

Since the two approaches do not give the same answer, the limit does not exist.

(c)

(a) $\dfrac{\partial u}{\partial x} = (e^{-x})(\sin y) + (-e^{-x})(x \sin y - y \cos y) = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y) = -2e^{-x} \sin y + xe^{-x} \sin y - ye^{-x} \cos y \qquad (1)$$

$$\frac{\partial u}{\partial y} = e^{-x}(x \cos y + y \sin y - \cos y) = xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}(xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y) = -xe^{-x} \sin y + 2e^{-x} \sin y + ye^{-x} \cos y \qquad (2)$$

Adding (1) and (2) yields $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ and u is harmonic.

Answer to the question no 7

(b)

5.7 Evaluate $\int_C \frac{3z^2+z}{z^2-1}\,dz$, where C is the circle $|z-1|=1$.

Solution The integrand has singularities, where $z^2-1=0$ i.e. at $z=1$ and $z=-1$.
The circle $|z-1|=1$ has center at $z=1$, $f(z)=3z^2+z$, is an analytic function.

Also,
$$\frac{1}{z^2-1}=\frac{1}{(z-1)(z+1)}=\frac{1}{2}\left(\frac{1}{z-1}-\frac{1}{z+1}\right)$$

$\therefore$
$$\int_C \frac{3z^2+z}{z^2-1}dz=\frac{1}{2}\int_C \frac{3z^2+z}{z-1}\,dz-\frac{1}{2}\int_C \frac{3z^2+z}{z+1}\,dz$$

By Cauchy's integral formula,

$$\int_C \frac{3z^2+z}{z-1}dz=2\pi i f(1)=8\pi i \qquad \text{where } f(z)=3z^2$$

By Cauchy's theorem, $\int_C \frac{3z^2+z}{z+1}\,dz=0$

$\therefore$ From (1), we have $\int_C \frac{3z^2+z}{z^2-1}\,dz=4\pi i$.

(c)

6.2 (a) Prove that the series $z(1-z)+z^2(1-z)+z^3(1-z)+\ldots$ converges for $|z|<1$, and (b) find its sum.

Solution The sum of the first n terms of the series is

$$\begin{aligned} S_n(z) &= z(1-z)+z^2(1-z)+\ldots+z^n(1-z) \\ &= z-z^2+z^2-z^3+\ldots+z^n-z^{n+1} \\ &= z-z^{n+1} \end{aligned}$$

Now $|S_n(z)-z|=|-z^{n+1}|=|z|^{n+1}<\epsilon$ for $(n+1)\ln|z|<\ln\epsilon$, i.e. $n+1>\frac{\ln\epsilon}{\ln|z|}$ or $n>\frac{\ln\epsilon}{\ln|z|}-1$ if $z\neq 0$.

If $z=0$, $S_n(0)=0$ and $|S_n(0)-0|<\varepsilon$ for all n.

Hence $\lim_{n\to\infty} S_n(z)=z$, the required sum for all z such that $|z|<1$.

Another method.

Since $S_n(z)=z-z^{n+1}$, we have [by Problem 2.41, Chapter 2. in which we showed that $\lim_{n\to\infty} z^n=0$ if $|z|<1$]

$$\text{Required sum} = S(z)=\lim_{n\to\infty} S_n(z)=\lim_{n\to\infty}(z-z^{n+1})=z$$

Answer to the question no 8

(a)

ndicated.

1. **Poles.** If $f(z)$ has the form (6.8) in which the principal part has only a finite number of terms given by

$$\frac{a_{-1}}{z-a}+\frac{a_{-2}}{(z-a)^2}+\cdots+\frac{a_{-n}}{(z-a)^n}$$

where $a_{-n} \neq 0$, then $z = a$ is called a *pole of order n*. If $n = 1$, it is called a *simple pole*.

If $f(z)$ has a pole at $z = a$, then $\lim f(z) = \infty$ [see Problem 6.32].

ENTIRE FUNCTIONS

A function which is analytic everywhere in the finite plane [i.e. everywhere except at ∞] is called an *entire function* or *integral function*. The functions e^z, $\sin z$, $\cos z$ are entire functions.

An entire function can be represented by a Taylor series which has an infinite radius of convergence. Conversely if a power series has an infinite radius of convergence, it represents an entire function.

Note that by Liouville's theorem [Chapter 5, Page 5.2] a function which is analytic *everywhere including* ∞ must be a constant.

MEROMORPHIC FUNCTIONS

A function that is analytic everywhere in the finite plane except at a finite number of poles is called a *meromorphic function*.

Example 6.5 $\dfrac{z}{(z-1)(z+3)^2}$ which is analytic everywhere in the finite plane except at the poles $z = 1$ (simple pole) and $z = -3$ (pole of order two) is a meromorphic function.